

Ugochukwu Okonkwo

Chester, PA (Open to Relocate) | 267-968-3159 |

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PROFESSIONAL SUMMARY

Biomedical Engineer with dual Master's degrees in Biomedical Engineering and Management and hands-on experience in medical device design, CAD modeling, prototyping, usability testing, and regulated product development. Experienced in supporting orthopedic implant and surgical instrument development, DV&V activities, engineering documentation, and cross-functional collaboration within MedTech environments. Combines technical engineering expertise with business and project management knowledge to support the development of innovative, patient-centered medical devices.

RELEVANT WORK EXPERIENCE

Globus Medical, Audubon, PA

Project Engineer Co-op, Jan 2024 - Aug 2024

- Applied Creo Parametric and engineering drawing standards to support development of orthopedic implants and surgical instruments in a regulated MedTech environment
- Utilized rapid prototyping, DFM principles, and cross-functional collaboration to support design iterations, manufacturability improvements, and functional evaluations
- Supported DV&V activities, engineering change processes, and technical documentation aligned with FDA and MDR quality system requirements
- Developed Excel macro-based inspection and traceability tools to improve quality documentation efficiency and workflow standardization
- Coordinated with manufacturing, operations, prototype shops, and suppliers to support prototype builds, testing, and production readiness
- Applied product lifecycle management concepts including DCO processing, UDI tracking, and DHF/DMR documentation support

EDUCATION

Widener University, Chester, PA

Master of Science in Engineering in Biomedical Engineering (Thesis), May 2026

Thesis: Ergonomic Redesign of an Assistive Button Hook for Individuals with Dexterity Impairments

Master of Business Administration in Management, May 2026

Bachelor of Science in Biomedical Engineering, May 2025

Minors: Management, Mathematics, Mechanical & Robotics Engineering

RESEARCH EXPERIENCE

Master's Thesis Research

Widener University | Aug 2025 – Present

- Applied human-centered design, ergonomics, and usability engineering principles to redesign an assistive dressing device for individuals with dexterity impairments
- Utilized CAD modeling, iterative prototyping, and 3D printing to develop and refine multiple assistive device concepts and handle geometries
- Designed and fabricated custom wire-forming jigs using CO₂ laser cutting and drill press fabrication techniques to improve repeatability during prototype development
- Conducted usability benchmarking, user feedback analysis, and comparative device evaluations to guide design optimization and performance improvements
- Collaborated with PT/OT stakeholders to integrate clinical and ergonomic feedback into engineering design decisions
- Applied prototype fabrication, mechanical assembly, and wire-forming techniques during development of assistive device mechanisms and functional testing

Low-Cost Versatile Electrospinning System

Undergraduate Senior Research Project, Widener University | Aug 2024 – April 2025

- Applied CAD modeling, prototype fabrication, and mechanical assembly techniques to develop a low-cost electrospinning platform for nanofiber production
- Utilized bandsaws, table saws, and fabrication tools to machine acrylic panels and aluminum structural components for custom system integration
- Performed aluminum tapping, fastening, and enclosure assembly operations to support structural stability and modular component integration
- Incorporated iterative prototyping, 3D printing, and system integration methods to refine syringe pump and rotating collector assemblies

- Conducted experimental parameter optimization involving flow rate, collector speed, and polymer delivery to improve fiber alignment and consistency
- Implemented safety-focused engineering controls including insulated acrylic housing, grounding, and controlled high-voltage operation

ADDITIONAL WORK EXPERIENCE

Widener University Office of Student Success, Chester, PA

Graduate Student Employee, Aug 2025 - May 2026

- Coordinated tutor-student matching and supported front desk operations, improving service efficiency, scheduling, and student support delivery.
- Trained and onboarded new tutors, reinforcing academic support protocols and effective communication strategies.
- Analyzed tutoring and engagement data using Microsoft Excel (pivot tables, queries) and developed promotional materials to increase program effectiveness and student participation.

Peer & Math Tutor, Sep 2024 - May 2025

- Delivered one-on-one and group tutoring across courses (MGT-210, MATH-242, ENGR-213), adapting teaching strategies to diverse learning styles.
- Simplified complex mathematical and technical concepts while guiding students through problem-solving, projects, and exam preparation.
- Fostered student confidence, independent learning, and problem-solving skills.

Widener University Office of Admissions, Chester, PA

Crew Leader, Aug 2023 - Dec 2023

- Led orientation programs and delivered presentations to groups of 20+ students, supporting their transition into university life.
- Collaborated with admissions and student affairs teams during university events and outreach initiatives

CERTIFICATIONS

ISO 14971: Medical Device Risk Management

Medical Device Process Validation (ISO 13485, IQ/OQ/PQ)

Geometric Dimensioning & Tolerancing (GD&T): Advanced Concepts

Lean Six Sigma Green Belt

TECHNICAL SKILLS

- CAD & Design: Creo Parametric, Autodesk Inventor, SolidWorks, 3D Printing, Iterative Design, Prototyping
- Modeling & Simulation: ANSYS Fluent, Thermal & Bioheat Modeling, Mesh Independence Studies
- Data & Analysis: MATLAB, Excel (Regression, Pivot Tables, Data Analysis ToolPak), SQL, Tableau
- Engineering & Medical Device: Design Verification & Validation (V&V), Human-Centered Design, Ergonomics, Usability Testing, Quality Systems (QMS), CO₂ Laser Cutting, Bandsaw & Table Saw Operation
- Quality & Regulatory: Lean Six Sigma, DMAIC, FMEA, Process Improvement, Design Controls
- Laboratory Techniques: Bradford Assay, Protein Extraction (RIPA Buffer)

PROFESSIONAL SKILLS

Research communication • Technical writing • Project management • Mentorship • Leadership • Collaboration & Teamwork • Problem-solving & Critical thinking